

Retail Order Management & Refund Process Improvement

1. Executive Summary

Nova Retail Pvt. Ltd. manages customer orders through both physical stores and its online sales channel. As order volumes increased, individual departments started maintaining separate Excel files to manage order processing, returns, refunds, and operational reporting.

This created inconsistencies between departments, increased manual reporting effort, and made it difficult to track the complete lifecycle of an order.

The objective of this project is to design a centralized reporting solution that consolidates operational data into a single source of truth. The proposed solution enables business teams to monitor orders, returns, refunds, and key operational KPIs through interactive dashboards, reducing manual effort and improving visibility across departments.

2. Business Background

Nova Retail Pvt. Ltd. is a mid-sized retailer selling consumer electronics through multiple sales channels. The company processes approximately 200,000 customer orders every year involving several business functions, including Operations, Warehouse, Customer Support, and Finance.

Each department performs its own responsibilities effectively but maintains separate operational trackers. Over time, this has resulted in duplicate information, inconsistent reporting, and delays in obtaining accurate business insights.

Managers currently spend significant time combining data from multiple sources before preparing daily or weekly operational reports. As the business continues to grow, this approach is becoming increasingly difficult to maintain.

3. Problem Statement

The current order management process is dependent on multiple Excel workbooks maintained independently by different departments.

This creates several operational challenges:

- No centralized view of order status.
- Returns and refunds are tracked separately.
- Reports are prepared manually.
- Different departments often work with different versions of the same data.
- Pending refunds are difficult to monitor.
- Operational bottlenecks are identified only after manual analysis.

These issues increase reporting effort, reduce operational visibility, and delay business decision-making.

4. Business Objectives

The proposed solution aims to:

- Create a centralized reporting solution for order management.
- Provide complete visibility into the order lifecycle.
- Improve tracking of returns and refunds.
- Reduce manual effort involved in operational reporting.
- Monitor business KPIs through interactive dashboards.
- Support faster operational decision-making.
- Improve collaboration between Operations, Warehouse, Customer Support, and Finance.

5. Stakeholder Analysis

Stakeholder	Role	Responsibility
Customer Support	Business User	Handle customer enquiries, return requests and refund follow-ups.
Operations Team	Business User	Manages order processing and order status updates.
Warehouse Team	Business User	Handles dispatch, delivery updates and returned products.
Finance Team	Business User	Processes customer refunds after verification.
Business Manager	Decision Maker	Reviews KPIs, monitors operations and identifies performance issues.
Business Analyst	Project Team	Gathers requirements, documents business processes and designs reporting solution.

6. Scope

In Scope

- Order tracking
- Return management
- Refund monitoring
- Operational KPI reporting
- Dashboard development
- Process performance analysis

Out of Scope

- Payment gateway integration
- Inventory management
- Customer-facing website
- ERP implementation

7. Current Process (AS-IS)

The existing process relies on multiple Excel trackers maintained independently by each department.

A typical workflow is as follows:

Customer places an order.

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Operations record the order.

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Warehouse dispatches the product.

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Customer receives the product.

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If required, the customer requests a return.

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Customer Support records the return request.

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Warehouse verifies the returned product.

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Finance processes the refund.

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Managers manually combine multiple reports to prepare business dashboards.

Because each department maintains separate records, considerable manual effort is required to produce accurate operational reports.

8. Key Business Challenges

Issue	Business Impact
Multiple Excel files	Duplicate information across departments
Manual reporting	Increased reporting time
Delayed refund tracking	Poor customer experience
Inconsistent order status	Reduced operational visibility
No centralized dashboard	Difficult KPI monitoring
Manual reconciliation	Increased chances of reporting errors

9. Gap Analysis

The analysis of the current business process identified several gaps between the existing way of working and the desired future state. Addressing these gaps will improve operational efficiency, reduce manual effort, and provide better visibility across departments.

Current State	Future State	Business Benefit
Each department maintains separate Excel trackers.	A centralized reporting solution stores all operational data in one place.	Eliminates duplicate information and improves consistency.
Reports are prepared manually.	Dashboards update automatically using a single data source.	Saves reporting time and improves accuracy.
Order, return and refund information is spread across multiple files.	Complete order lifecycle is available from a single dashboard.	Better visibility across departments.
Managers identify issues after manual review.	KPIs and pending cases are available in real time.	Faster decision making.
Refund tracking requires manual follow-up.	Pending refunds are highlighted automatically.	Improves customer satisfaction.

11. Functional Requirements

ID	Requirement	Priority
FR-01	The system shall maintain a centralized record of user orders.	High
FR-02	The system shall track the complete lifecycle of every order.	High
FR-03	The system shall record customer return requests.	High
FR-04	The system shall track refund status until completion.	High
FR-05	Users shall be able to search orders using Order ID /Customer ID.	Medium
FR-06	Managers shall be able to view operational dashboards.	High
FR-07	Dashboard shall display pending refunds.	High
FR-08	Dashboard shall display delayed orders.	High
FR-09	Dashboard shall display operational KPIs.	High
FR-10	Reports shall support filtering by date and department.	Medium
FR-11	Users shall be able to export reports to Excel or PDF.	Medium
FR-12	Dashboard shall refresh whenever new operational data becomes available.	Medium

12. Non-Functional Requirements

ID	Requirement	Description
NFR-01	Performance	Dashboard should load within five seconds under normal usage.
NFR-02	Availability	Reporting solution should be available during business hours.
NFR-03	Reliability	Reports should accurately reflect available business data.
NFR-04	Security	Access should be restricted based on user roles.
NFR-05	Scalability	Solution should support increasing order volumes.
NFR-06	Usability	Dashboard should be simple and easy to navigate.
NFR-07	Maintainability	New KPIs and reports should be added with minimal effort.

13. Business Rules

The following business rules will govern the reporting solution:

- Every order must have a unique Order ID.
- Every return request must be linked to an existing order.
- Refunds can only be processed after return verification.
- Refund status can only be updated by the Finance team.
- Order status can only be updated by the Operations team.
- Shipment status can only be updated by the Warehouse team.
- Dashboard metrics should always be generated from the latest available operational data.

14. Data Requirements

The reporting solution requires operational data from multiple business functions.

Data Entity	Description
Orders	Customer order information including order status.
Customers	Customer details required for reporting.
Returns	Information related to returned products.
Refunds	Refund requests and payment status.
Warehouse	Shipment and return verification details.
Operations	Order processing activities.

The solution assumes that operational data is available in a structured format before it is loaded into the reporting system.

15. Reporting Requirements

The reporting solution should provide dashboards that enable different business users to monitor operational performance.

Executive Dashboard

Displays:

- Total Orders
 - Completed Orders
 - Pending Orders
 - Returned Orders
 - Refunded Orders
 - Overall operational KPIs
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Operations Dashboard

Displays:

- Orders processed
 - Pending orders
 - Delayed orders
 - Processing time
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Warehouse Dashboard

Displays:

- Orders awaiting shipment
 - Products dispatched
 - Returned products awaiting verification
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Finance Dashboard

Displays:

- Pending refunds
- Completed refunds
- Average refund processing time
- Refund trend

Customer Support Dashboard

Displays:

- Open customer cases
- Return requests
- Pending customer complaints
- Resolution status

16. Success Metrics

The success of the proposed solution will be measured using the following business metrics:

KPI	Expected Improvement
Manual report preparation time	Reduced
Pending refund visibility	Improved
Cross-department reporting	Centralized
Dashboard availability	Improved
Operational reporting accuracy	Improved
Time required to identify delayed orders	Reduced

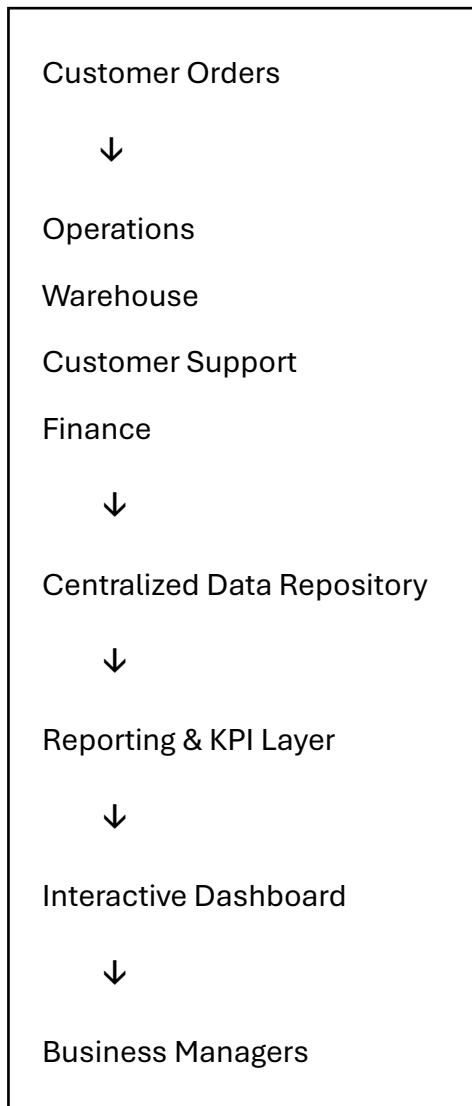
17. High-Level Solution Overview

The proposed solution follows a centralized reporting architecture.

Business data generated by Operations, Warehouse, Customer Support, and Finance is collected into a common data repository.

The reporting layer processes this information and presents business insights through interactive dashboards, allowing management to monitor operational performance without manually consolidating multiple Excel files.

Solution Flow:



18. Acceptance Criteria

The project will be considered successful when:

- Users can view the complete order lifecycle from a single dashboard.
- Managers can monitor operational KPIs without manual report preparation.
- Pending returns and refunds are clearly identified.
- Reports can be filtered based on business requirements.
- Dashboard information accurately reflects operational data.
- Business users can generate reports without assistance from technical teams.

19. Appendix

Term	Description
KPI	Key Performance Indicator
Order Lifecycle	Complete journey of an order from placement to refund completion
Dashboard	Visual reporting interface used for monitoring business performance
Refund	Amount returned to the customer after an approved return

References

- Business Process Discussions
- Operational Reporting Requirements
- Existing Excel Tracking Process

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